DevOrbis_Final — Test Plan

Document Type: QA / SQA Test Plan
Project: DevOrbis_Final — AI Operations Hub
Application URL: https://dev-orbis-final-frontend.vercel.app/
Baseline Document: DevOrbis_Final Requirement Analysis Report v1.0
Document Version: 1.0
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Test Approach: Black-box functional and non-functional testing
Planned Test Level: System / End-to-End web application testing
Test Status: Planned
Date: 4 October 2026

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1. Test Plan Purpose

This Test Plan defines the scope, objectives, approach, resources, environments, entry/exit criteria, risks, and deliverables for testing DevOrbis_Final — AI Operations Hub.

It is derived from the Requirement Analysis Report v1.0 and the Software Requirements Specification (SRS). This document is intentionally limited to test planning; detailed test scenarios and test cases will be produced as separate QA artifacts.

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2. Test Objectives

- Verify that the implemented functionality conforms to approved functional requirements.
- Verify authentication, authorization, and role-based access across User, Staff, and Admin roles.
- Verify chat infrastructure (AI Text Chat and AI Voice Chat).
- Verify RAG pipeline behaviour including retrieval, reranking, grounding, and citation.
- Verify lead management, AI lead qualification, and lead-task association.
- Verify task management (create, assign, update, status transitions).
- Verify AI-assisted task creation with human-in-the-loop review.

- Verify dashboards and AI Usage metric accuracy.
- Verify integrations (n8n webhook, Monday.com sync).
- Assess non-functional aspects (performance, security, reliability, usability) within agreed scope.
- Identify, document, retest, and regress defects found during execution.
- Maintain traceability between requirements, scenarios, and test cases.

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3. Test Scope

3.1 In Scope

Module / Area	Planned Coverage
Landing Page	Public site, navigation, CTAs
Authentication	Sign Up, Login, Email Verification, Logout, Session
Authorization	Role-based access (User / Staff / Admin)
User Role	Home, AI Text Chat, AI Voice Chat, Logout
Staff Role	Staff Dashboard, My Tasks, Create Task
Admin Role	Admin Dashboard, Knowledge Base, All Tasks, Task Detail, Edit Task, AI Usage, AI Conversations, Create Task
Chat	New chat, previous chats, message persistence, streaming/real-time
AI Assistant	Response generation, error handling, out-of-scope queries
Knowledge Base	Upload, list, download, delete documents
RAG	Retrieval, reranking, grounding, citation
Lead Management	Lead list, lead detail, lead lifecycle
AI Lead Qualification	Qualification trigger, lead creation
Task Management	Create, assign, update, status transitions

Module / Area	Planned Coverage
AI-Assisted Task Creation	Natural language input, generated fields, editable form
Dashboards	Admin metrics, Staff metrics, Task Overview
AI Usage	Tokens, cost, refresh behaviour
Integrations	n8n webhook, Monday.com sync
Voice AI	Voice Chat interaction, STT, TTS, tool calls
Role-Based Navigation	Sidebar visibility and access enforcement
UI/UX	Labels, buttons, messages, consistency, responsiveness, loading/ error/empty states
Non-Functional	Performance, security, reliability, usability, compatibility

3.2 Out of Scope / Not Yet Baseline

- Backend internals, database schema, or API implementation testing without approved technical requirements.
- AI model training, embedding configuration, or reranker tuning beyond product-level behaviour.
- Third-party Monday.com internals; only application-side integration behaviour is in scope.
- n8n workflow internals; only webhook trigger and payload verification is in scope.
- Performance thresholds and security acceptance levels that have not yet been defined.
- Penetration testing or formal security audit.
- Legal or compliance certification testing.

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4. Test Items

ID	Test Item
TI-01	Landing Page

ID	Test Item
TI-02	Authentication (Sign Up, Login, Verification, Logout)
TI-03	Authorization and Role-Based Access
TI-04	User Home
TI-05	AI Text Chat
TI-06	AI Voice Chat
TI-07	Staff Dashboard
TI-08	My Tasks
TI-09	Create Task
TI-10	Admin Dashboard
TI-11	Knowledge Base
TI-12	All Tasks
TI-13	Task Detail
TI-14	Edit Task
TI-15	AI Usage
TI-16	AI Conversations
TI-17	Lead Management
TI-18	AI Lead Qualification
TI-19	RAG Pipeline
TI-20	AI-Assisted Task Creation
TI-21	Integrations (n8n / Monday.com)
TI-22	Non-Functional Quality (Performance, Security, Reliability, Usability)

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5. Test Strategy

5.1 Functional Testing

Verify each approved requirement behaves according to its expected result. Includes positive, negative, boundary, and validation checks.

5.2 Integration / End-to-End Testing

Validate connected user journeys such as:

- Sign Up → Verify → Login → Chat → Lead qualification
- Login (Staff) → View Dashboard → Create Task
- Login (Admin) → Upload Document → Chat → RAG Retrieval
- Admin → Create Task → Assign Staff → Sync to Monday.com

5.3 Regression Testing

After defect fixes or material changes, impacted functionality and related areas will be re-executed.

5.4 UI / Usability Testing

Verify visibility, labels, consistency, action discoverability, validation messages, and responsive behaviour across supported viewports.

5.5 AI / RAG Testing

AI testing focuses on application-level behaviour:

- Retrieval relevance
- Reranking effectiveness
- Grounding of responses
- Citation correctness
- Hallucination handling
- Behaviour when no relevant chunks retrieved

5.6 AI Security Testing

- Prompt injection attempts
- Malicious document content
- Data leakage through AI responses
- Authorization boundaries maintained independently of AI

5.7 Voice AI Testing

- Voice interaction initiation
- STT accuracy (transcription)
- TTS output (pronunciation, pacing, naturalness)
- Tool calls (if applicable)

- Business outcomes (task creation, lead updates)
- Post-call verification

5.8 Security-Oriented Functional Checks

Within SQA scope: unauthorized access attempts, session behaviour, input validation, exposure of data through normal user flows. Penetration testing is out of scope unless separately authorized.

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6. Test Types

Test Type	Purpose	Planned
Smoke Testing	Confirm build is stable enough for deeper testing	Yes
Functional Testing	Verify approved functional requirements	Yes
Negative Testing	Verify invalid/missing/ unsupported input handling	Yes
Boundary Testing	Validate defined input limits and edge conditions	Yes
Integration / E2E	Validate connected workflows	Yes
Regression Testing	Verify fixes and change impact	Yes
UI / Usability	Evaluate interface behaviour and clarity	Yes
Compatibility	Verify supported browsers and viewports	Yes
Performance	Measure response behaviour against agreed thresholds	Yes, after thresholds are defined
Security Checks	Verify basic access/ session protections	Yes
AI / RAG Behaviour	Validate product-level AI workflow behaviour	Yes

Test Type	Purpose	Planned
AI Security	Prompt injection, data leakage checks	Yes
Voice AI	Test voice chat interaction end-to-end	Yes
Accessibility	Basic accessibility observations	Yes

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7. Test Environment

Component	Planned Environment / Requirement
Application	Deployed DevOrbis_Final on Vercel
Backend	Deployed Express backend on Railway
Database	Neon PostgreSQL
Cache	Redis Cloud (production) / redis://localhost:6379 (local)
Browser	Google Chrome (primary); additional supported browsers after confirmation
Operating System	Windows (baseline); other supported OS as required
Viewport	Desktop baseline plus responsive widths (mobile, tablet)
Network	Stable internet; network degradation used for resilience checks
Test Data	Dedicated test accounts for User, Staff, Admin; sample documents for KB; sample leads and tasks
AI Dependency	Gemini (embeddings + LLM) available
Integration Dependency	n8n instance and Monday.com account configured
Voice Dependency	Voice AI service configured (if applicable)

Credentials and production-sensitive data should not be embedded in test evidence. Test data should be controlled and resettable.

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8. Entry Criteria

- Application build/deployment is accessible and reasonably stable.
- Approved or mentor-confirmed requirements are available for functionality under test.
- Test environment and required dependencies are available.
- Test accounts for all three roles (User, Staff, Admin) are available.
- Required test data (documents, leads, tasks) is available.
- Known blockers preventing execution are documented.
- For AI testing, AI services are available.
- For integration testing, n8n and Monday.com are configured.
- For voice testing, voice service is available or an agreed test mode exists.

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9. Exit Criteria

- All planned test cases for the agreed scope have been executed or formally deferred.
- All critical/high-severity defects are resolved, retested, or formally accepted.
- Regression testing completed for resolved defects and impacted areas.
- Requirement-to-test traceability is complete for approved scope.
- Test execution results and defect status are documented.
- Remaining risks and unverified requirements are communicated to the mentor/product owner.

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10. Suspension / Resumption Criteria

10.1 Suspend Testing

- Application unavailable or blocking environment failure.
- Critical defect prevents execution of a major workflow.
- Required test data or external dependency unavailable (AI, n8n, Monday.com, Redis).
- Major build change invalidates current test baseline.

10.2 Resume Testing

- Environment/build restored and smoke-tested.
- Blocking defects/dependencies resolved or approved workaround available.
- Changed requirements or build behaviour reviewed and affected coverage updated.

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11. Defect Management Approach

Defects will be recorded with:

- Unique ID
- Concise title
- Module
- Environment
- Preconditions
- Steps to reproduce
- Expected result
- Actual result
- Severity and Priority
- Evidence (screenshots, logs, trace IDs)
- Failure stage attribution (for AI / RAG defects: ingestion, chunking, retrieval, context, generation, tool selection, tool execution, post-processing)
- Status

Fixed defects will be retested and related regression coverage executed.

Severity Definitions

Severity	Meaning
Critical	Application or core workflow unusable; major data/security impact; no viable workaround
High	Major function fails and materially blocks a primary workflow
Medium	Functional issue with workaround or limited impact

Severity	Meaning
Low	Minor UI/content/usability issue with limited functional impact

Priority describes how urgently the team should address the defect; it is independent of severity.

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12. Roles and Responsibilities

Role	Responsibilities
SQA Intern / Tester	Requirement review, test design, execution, evidence collection, defect reporting, retesting, regression testing
Mentor / Product Owner	Clarify requirements, confirm expected behaviour, prioritize/accept defects, approve ambiguous business rules
Developer	Investigate and fix defects, provide build updates and technical clarification
Project / Team Lead	Coordinate releases, scope changes, final test status/acceptance decisions

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13. Risks and Mitigation

Risk	Impact	Mitigation
Incomplete requirements	Incorrect or incomplete test coverage	Confirm TBD items before final test design
AI output variability	Results vary between runs	Define product-level acceptance criteria and repeatability approach
AI service outage	Blocks chat/RAG/lead qualification	Record dependency availability; use controlled failure tests
n8n / Monday.com outage	Blocks integration tests	Record dependency availability; validate retry behaviour

Risk	Impact	Mitigation
Redis unavailable	Blocks cached metric verification	Use fallback checks; verify error handling
Insufficient test data	Limits positive/negative coverage	Create dedicated reusable accounts, documents, leads, tasks
Frequent deployments	Invalidates results	Record build/version per cycle; smoke/regression after changes
Voice AI service availability	Blocks voice testing	Confirm service availability before voice test cycle
Browser/device assumptions	Compatibility conclusions unreliable	Obtain official support matrix before sign-off

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14. Test Deliverables

- Requirement Analysis Report
- Approved Test Plan
- Test Scenario document
- Detailed Test Case document (regular + Voice AI)
- Test execution results / evidence
- Defect / Bug Report document
- Retest and regression results
- Requirement Traceability Matrix (RTM)
- Final Test Summary Report

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15. Requirement-to-Test Coverage Plan

Requirement Area	Planned Coverage
Authentication	Registration, login, verification, logout, session
Authorization	Role-based access, cross-role URL access, backend enforcement
User Role	Home, Chat, Voice Chat, Logout
Staff Role	Dashboard, My Tasks, Create Task

Requirement Area	Planned Coverage
Admin Role	Dashboard, Knowledge Base, All Tasks, Task Detail, Edit Task, AI Usage, AI Conversations, Create Task
Chat	New chat, previous chats, message flow, real-time events
AI Assistant	Response generation, error handling, out-of-scope queries
Knowledge Base	Upload, list, download, delete, error handling
RAG	Retrieval, reranking, grounding, citation, empty results
Lead Management	Lead list, lead detail, uniqueness, status
AI Lead Qualification	Trigger, record creation, priority
Task Management	Create, assign, update, status transitions, validation
AI-Assisted Task Creation	Input, generation, review, submission
Dashboards	Metrics accuracy, caching behaviour
AI Usage	Tokens, cost, refresh
Integrations	Webhook trigger, payload, failure handling, sync behaviour
Voice AI	Voice initiation, STT, TTS, tool calls, outcomes, post-call verification
Role-Based Navigation	Sidebar visibility, direct URL access behaviour
UI	Consistency, labels, loading/error/ empty states, responsiveness
Non-Functional	Performance, security, reliability, usability, compatibility

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16. Test Execution Metrics

Metric	Definition
Total Test Cases	Number of approved test cases in cycle

Metric	Definition
Executed	Number actually run
Passed	Executed tests meeting expected results
Failed	Executed tests not meeting expected results
Blocked	Tests that cannot execute due to environment/dependency/defect
Pass Rate	Passed ÷ Executed × 100
Defects Found	Total unique defects logged
Defect Closure	Resolved/retested ÷ Total logged
Requirement Coverage	Requirements linked to ≥1 test ÷ Approved requirements × 100
Failure Stage Attribution	Failures grouped by pipeline stage

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17. Test Schedule — Planned Sequence

Phase	Activity	Output
Phase 1	Requirement confirmation and scope freeze	Approved requirement baseline (v1.1+)
Phase 2	Test scenario design	Test Scenario document
Phase 3	Detailed test case design + RTM	Test Cases + RTM
Phase 4	Environment/test-data setup + smoke test	Ready-for-test confirmation
Phase 5	Functional and negative execution	Execution results + defects
Phase 6	AI / RAG / Voice / Integration / Compatibility / NFR checks	Additional results + defects
Phase 7	Retest + regression	Updated execution/ defect status

Phase	Activity	Output
Phase 8	Closure	Test summary and residual-risk report